



NRA GEO-HAZARD ASSESSMENT

Survey Data



Operational excellence for results that matter

Steve Revill (UNOPS) – 09 August 2017



SURVEY DATA - OUVREVIEW

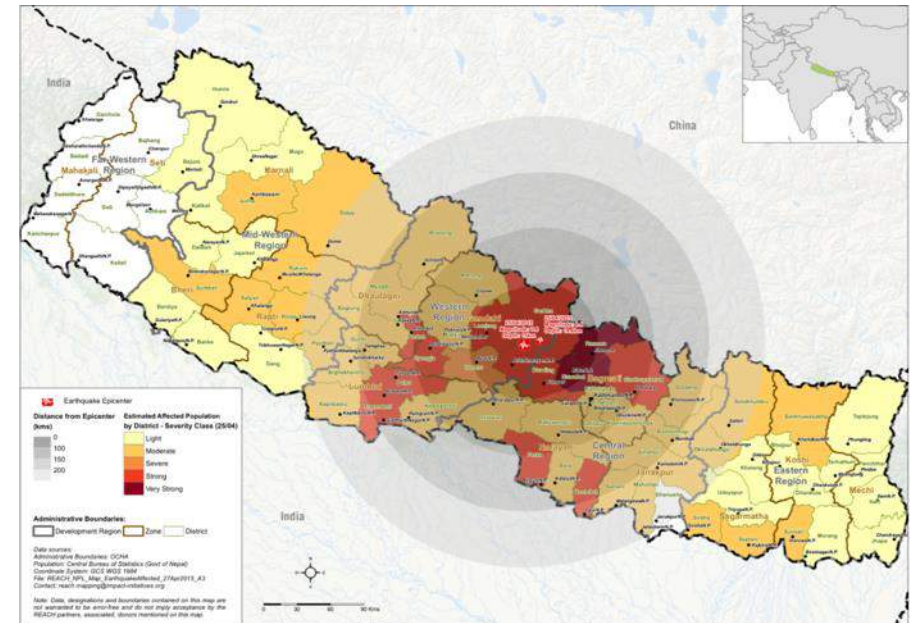


1. Background & Purpose of Survey
2. Survey Output – Data Formats
3. Field Review Examples
4. Next Steps + Technical Note on Remediation

1. BACKGROUND & PURPOSE OF SURVEY

- 2015 Ministry of Home Affairs Report identified locations at risk from geo-hazards following the Gorkha Earthquake
- Initial 130 surveys completed by the Department of Mines & Geology
- The start for this project: 455 locations in 15 Districts, this has increased since the survey started
- **Districts:** Ramchhap, Solukhumbu, Dhading, Dolakha, Makwanpur, Kavrepalanchok, Rasuwa, Chitwan, Gorkha, Sindhupalchowk, Okhaldunga, Lamjung, Tanahu, Kathmandu, Lalitpur

DISTRICT	Inspection Reference Code	HRRP_VNAME	WARD	SETTLEMENT	Complete	Outstanding	Easting	Northing
Nuwakot	DMG			Chhelun				
Dhading	525 2 05 30 5 100 Bo	Sertung		Borang	1		603843	3116000
Dhading	525 2 05 30 5 017 Jh	Jharlang	7	NECRANG TOLE	1		599631	3110324
Dhading	526 2 05 30 5 017 Jh	Jharlang	1	GALMRANG/DURCHA GAUN	1		601785	3112108
Dhading	527 2 05 30 5 017 Jh	Jharlang	1	CHYAMTHALI	1		601525	3111676
Dhading	528 2 05 30 5 017 Jh	Jharlang	7	SINGUN	1		280 05' 43"	850 01' 10"
Dhading	529 2 05 30 5 017 Jh	Jharlang	8	THALI	1		604804	3108384
Dhading	530 2 05 30 5 017 Jh	Jharlang	8	KIUNG	1		604577	3107691
Dhading	525 2 05 30 5 045 Se	Sertung	6	BERKAP	1		606687	3117489
Dhading	525 2 05 30 5 045 Se 9HI	Sertung	9	HINDUN	1		606157	3122182
Dhading	525 2 05 30 5 049 TI	Tipling	1	SIMPALE	1		607895	3119595
Dhading	525 2 05 30 5 049 TI 02 Th	Tipling	2	TULO GAUN	1		609040	3118538
Dhading	525 2 05 30 5 049 TI 08 Ph	Tipling	8	PHYANG GAU	1		609012	3119112
Dhading	DMG	Lapa		Chhamrang				
Dhading	DMG			Chapangchet				
Dhading	525 2 04 22 5 008 Dhn	Darkha	8	Dharna TOLE	1		597885	3106344
Dhading	DMG	ReeGaun						
Dhading	DMG			Gamrang				
Dhading	DMG			Richet				
Dhading	DMG			Taajimrang				
Dhading	DMG			Andar				
Dhading	DMG			Kichet				
Dhading	DMG			Cherprang				
Dhading	DMG			Kareng				
Dhading	DMG	Lapa						
Rasuwa	524 2 05 29 5 009 Ka	DHAIBUNG	2	Kadam ghaari	1		619173	3096522
Rasuwa	524 2 05 29 5 009 Dh	DHAIBUNG	3	Dhakaldanda	1		618835	3097551
Rasuwa	524 2 05 29 5 009 Gh	DHAIBUNG	7	Chitrapani	1		619956	3099224
Rasuwa	524 2 05 29 5 009 Pr	DHAIBUNG	7	PRANGSEY	1		619816	3099308
Rasuwa	524 2 05 29 5 001 Gh	BHORLE	1	GHATTE PAKHA	1		621492	3097880
Rasuwa	524 2 05 29 5 009 Ta	BHORLE	6	Tallo rusepani	1		621274	3096773
Rasuwa	524 2 05 29 5 003 Pa	Chilime	8,9	Pajung & Kerabari	1		85 19 55	28 10 59
Rasuwa	524 2 05 29 5 003 Ke	Chilime		Kerabari				
Rasuwa	524 2 05 29 5 003 Ga	Chilime	6	Ganggang	1		627724	3120162



1. BACKGROUND & PURPOSE OF SURVEY

- *Preliminary Assessment* to categorize communities considered to be at risk from earthquake induced geo-hazards (landslides).
- Classification of each community into one of three categories:
 - Category 1:** Safer communities/villages where reconstruction can be started any time.
 - Category 2:** Communities/villages under the risk of manageable geo-hazards. Reconstruction can be started only after applying the suitable countermeasures.
 - Category 3:** Unsafe communities/villages due to the existing state of geo-hazards which are extremely difficult to control technically as well as financially. Reconstruction is not recommended.
- Provide recommendations on:
 1. Approximation of number of households requiring relocation.
 2. Outline recommendations for remedial works and/or investigation.
 3. Identify responsible government agency for remediation.

Between the NRA & DMG Surveys almost 650 locations have been assessed

2. SURVEY OUTPUT



NRA GEO-HAZARD ASSESSEMENT SUMMARY SHEET

Settlement Reference Number	525204225008Dhn		
Settlement Name	Dhaxaa		
District	Dhading		
VDC & Ward Number	Dharkha-8		
GPS Coordinates	3106344N	597885E	Elevation:1010

1) Description of topography at location of settlement.
Dhaxaa Gaa is located on the middle of the hill slope facing NW with representative slope of 26°.

2) Brief description of geological conditions and engineering geology considerations.
The lithology of settlement area consists of Schist with representative orientation N295/24°SE. The Schist is highly weathered.
The curvature of slope of settlement is concave with presence of topographic break. The seepage condition around the settlement is dry. The soil type observed is colluviums. The stiffness of the soil mass is observed sticky.

3) Description of landslides affecting the settlement.
The landslide present in down slope is Earth flow with debris materials that has (120*35*1.5m) dimensions. At present, it is in suspended state. This shallow seated landslide was triggered by recent monsoon. Some tension cracks are observed at the crown. The slope of the landslide is very steep hence it has higher chances to reoccur in next monsoon. The landslide present at upslope is rock fall which is followed by Earth flow which lies within the forest zone. The main reason behind this landslide is rural road construction. One gabion wall is already constructed 25m below crown.

4) Number of houses and infrastructure affected.
The houses and infrastructure weren't affected by landslide but earthquake. The presence of landslide on the down slope can affect settlement (mainly 3 houses) in its future occurrence.

5) Proposed settlement category and recommendations for monitoring, investigation, remediation or relocation.
The Dhaxaa settlement is proposed Category III (Relocation needed for 4 houses) and also landslide control measures. Responsible Department: DSCWM

The following means of slope protection can be used:

- Surface water drainage
- Bioengineering
- Slope trimming followed by
- Retaining wall (Gabion wall)

The relocation site for 3 houses is proposed at
Private land relocates nearby in their land (around 200m away from landslide)

Name of Consultant	Kishor Acharya	Raj Kumar Eandel	Migmo Yangji Sherpa
Discipline	Geologist	SCWM	WIDM
Signature			
Date	15/01/2017	15/01/2017	15/01/2017



NRA GEO-HAZARD ASSESSEMENT

SETTLEMENT REFERENCE NUMBER:
255307365055 Ma

SETTLEMENT NAME: Mahabhir

DISTRICT: Gorkha

VDC & WARD: Simjung-1



Technical Committee Presentation
With Photographs
For each location

Main Report
For each location

A. Data Collection Sheet for Field Investigation
(Note: 1) As far as possible field photographs should be taken for each of the following items to illustrate the ground reality.
2) Walk over survey is to be conducted with GPS along the boundary of the landslide / settlements.

FORM 1 - SETTLEMENT INFORMATION

Primary Responsibility: SCWM

Settlement Reference Number..... 525204225008Dhn

District: Dhading, VDC: Darcha, Ward No.: 8,

Name of village: Dhaxaa,

Date: 15/01/2017,

Number of households in the settlement: 57,

Latitude: 3106344N, Longitude: 597885E, Elevation: 1010m,

Name(s) and discipline of consultants:

1) Kishor Acharya,

2) Raj Kumar Eandel,

3) Migmo Yangji Sherpa,

Position of the geo-hazards (tick the appropriate one): It is situated upslope/ down slope/ within / side slope of the settlement.

(Place a representative photograph and the topographical map of the settlement here)



1 Page
Summary Report
For each location

2. SURVEY OUTPUT

Summary tables published by NRA

INVITATION, HRRP and MoUD CLPIU Technical Session, Wednesday, 9 August, 2pm: this session will focus on the results of the geo-hazard survey conducted by NRA, and there will also be a presentation from the 'Supporting Durable Solutions' consortium regarding support for displaced families. The session will take place in the DUDBC Conference Hall.

1	2	3	4	5	6	7	8	9							
DISTRICT	Inspection Reference Code	HRRP_VNAME	WARD	SETTLEMENT	Vuln_HHs Complete	Outstanding	DMG Survey	Category 1	Category 2	Category 3	Survey Group	Easting	Northing	Responsible Agency	Remarks
Sindhupalchok		Gati		9 Gati Village	18		1	1			DMG				Safe from landslide. Dry cultivation and drainage management recommended.
Sindhupalchok	525 2 05 23 5 021 Ga	Gati		1 KHAGDOLE, Dalit Basti		1			1		Group 8	391780	3076906	DSCWM	Bioengineering, cascade, catch wall recommended
Sindhupalchok	524 2 05 23 5 029 Ya	Helumbu		9 Yamawalama (YAMBALAMA)	200	1		1			Group 6	357812	3094472	Community	Bioengineering works recommended to be undertaken by the community
Sindhupalchok	524 2 05 23 5 029 Am	Helumbu		8 Amawasthan (AMBATHAN)		1			1		Group 6	356762	3094437	DSCWM	Earthflow and rock fall require mitigation, crack sealing, bioengineering , block removal, slope trimming
Sindhupalchok	524 2 05 23 5 029 na	Helumbu		6 Nakote		1			1		Group 6	355863	3099369	DSCWM	Bioengineering, monitoring and removal of rock blocks recommended
Sindhupalchok	524 2 05 23 5 029 Gh	Helumbu		7 Ghaewaagaun	10	1				1	Group 6	356035	3096121	N/A	All 10 HOUSEHOLDS TO BE RELOCATED, NO SUITABLE LOCATION IDENTIFIED
Sindhupalchok	524 2 05 23 5 019 Ch	Fulpingkatti		8 Chaku bajar		1			1		Group 8	393314	3085401	DWIDM	Anchored retaining wall, bioengineering, drainage, sealing of cracks and monitoring are recommended.

1. District

2. VDC

3. Ward No.

4. Community Name

5. Survey Category
6. Survey Group

7. Geographic Location

8. Responsible Agency (DSCW, DWIDM, Community)

9. Brief summary of report conclusions/recommendations

NOTE THAT PUBLISHED DATA IS NOT THE LATEST VERSION

2. SURVEY OUTPUT

Summary tables for the examples described below:

- 1
- 2
- 3
- 4
- 5

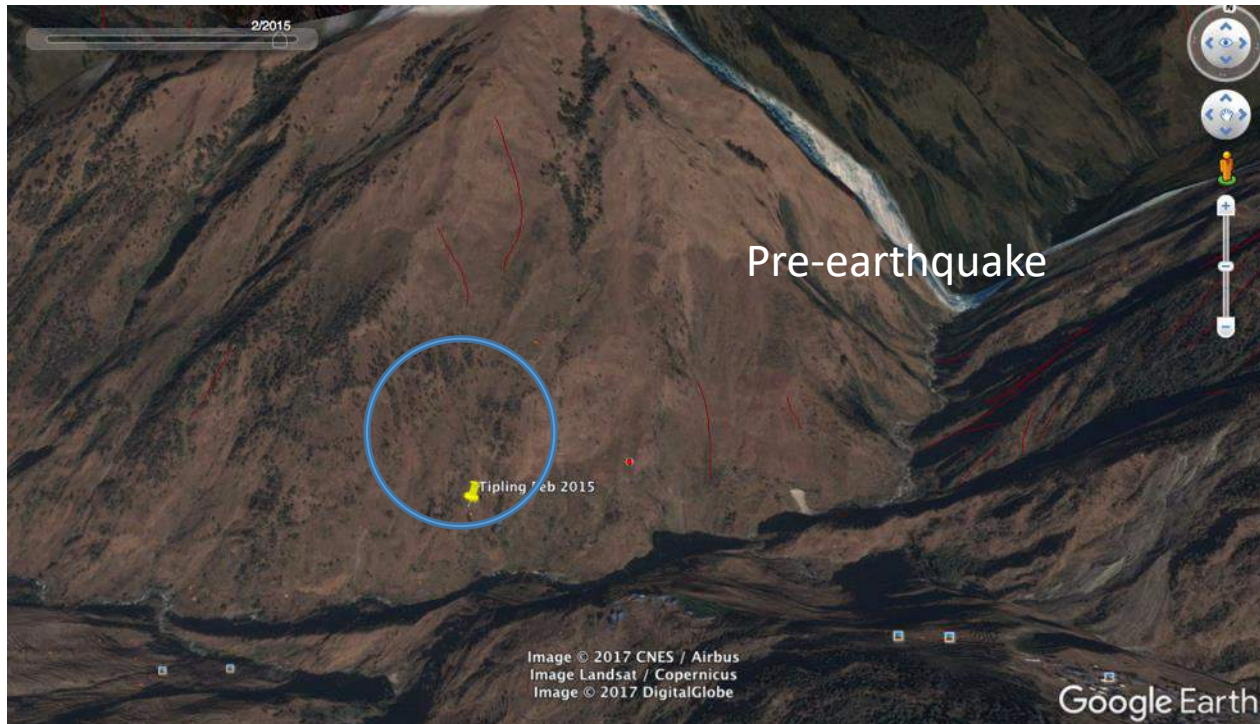
DISTRICT	HRRP_VNAME	WA	SETTLEMENT	Total Households in Community (PLEASE READ NOTE 1 below)	Total Affected Households (PLEASE READ NOTE 2 below)	Dor	Cal	Cat	Ca	Survey Gro	Easting	Northing	Responsible Agency	Remarks & Estimated Household Relocations (PLEASE READ NOTE 3 below)
Nuwakot	Gorsyang VDC		Kapre						1	DMG	604200	3089723		HOUSEHOLDS ALONG KHORUM KHOLA REQUIRE RELOCATION. Flood prevention measures are also recommended.
Dhading	Tipling	1	SIMPALE	18	18	1			1	Group 1	607895	3119595	N/A	ALL 18 HOUSEHOLDS TO BE RELOCATED
Rasuwa	Yarsa	5	Sangur bachha/SANGYUL	29	29	1			1	Group 3	628854	3102526	N/A	ALL 29 HOUSEHOLDS REQUIRE RELOCATION
Gorkha	Sirdibas	7	DEWAL GAUN	50	50	1		1		Group 14	588170	3141459	DSCWM	Surface drainage, river training and bioengineering recommended.
Sindhupalchok	Fulpingkatti	4	Hindi Luksing	80	55	1		1		Group 8	393866	3087719	DWIDM	Drainage, cascade construction, river training, monitoring

- 1 – Estimated total number of houses in the community
- 2 – Estimate of total number of houses in the community affected by landslides (NOT ALWAYS PROVIDED)
- 3 – Geo-Hazard Categorization
- 4 – Geographic location – two systems used
- 5 – Outline of recommendations and number of houses to be relocated where applicable.

3. FIELD REVIEW EXAMPLES

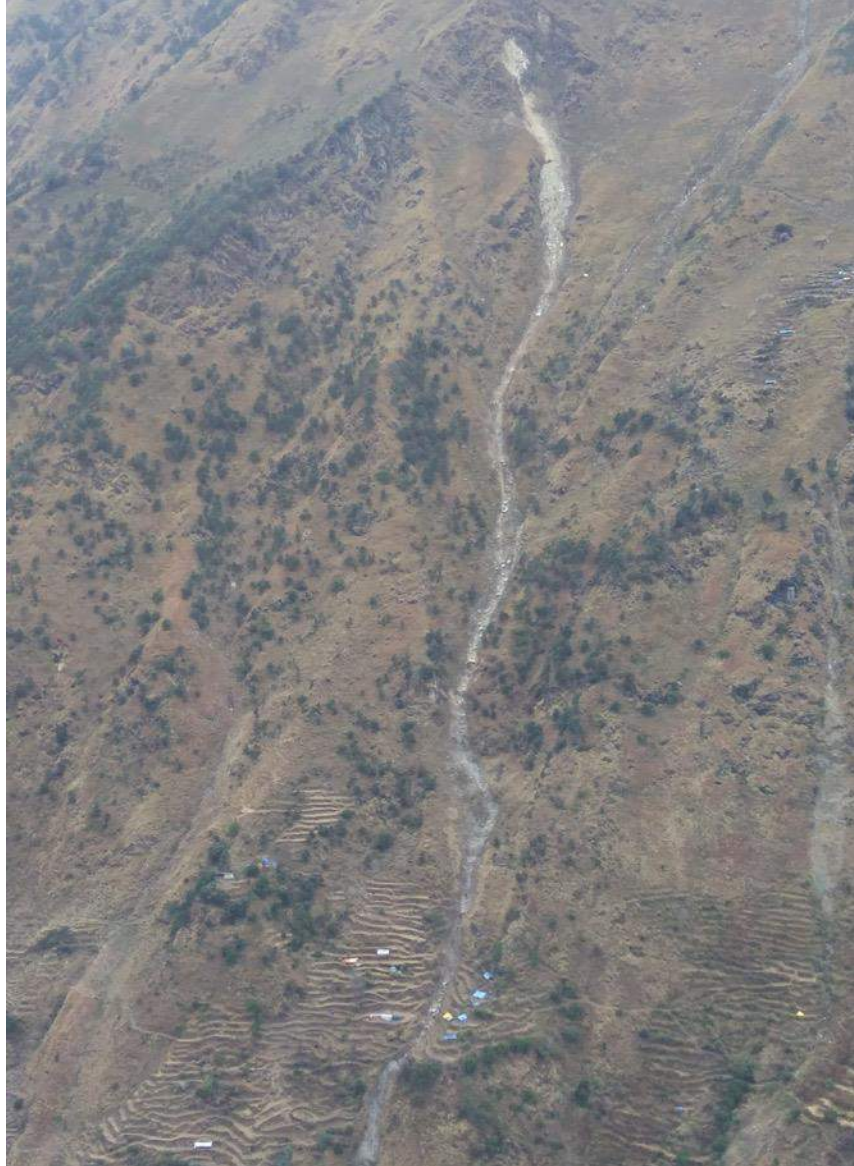
Dhading: Tipling VDC – Simpale

NRA ASSESSMENT – Category 3: All 18 households require relocation



LANDSLIDE DID NOT EXIST BEFORE THE EARTHQUAKE

Dhading: Tipling VDC – Simpale



1. Steep slope $>60^\circ$
2. Gully passes through the settlement
3. Material detachment area above settlement – debris source
4. Houses are $<5\text{m}$ from gully.
5. Erosion & debris flow prominent during monsoon.
6. A heavy monsoon could cause severe debris flow through the settlement.
7. Risk increases if the gully expands/changes direction.

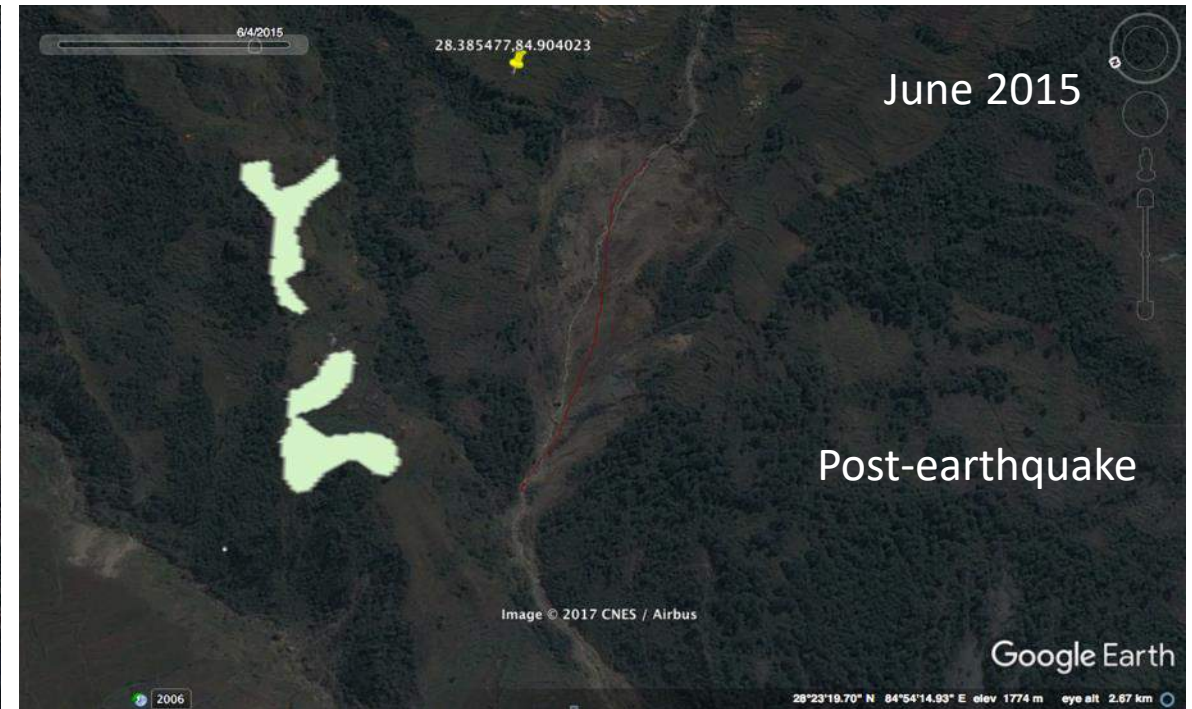
DURING RAINY SEASON GROUNDWATER PRESSURES WILL RESULT IN MORE RELEASE OF MATERIAL FROM SOURCE AREA GRAVITY WILL DO THE REST.

Dhading: Tipling VDC – Simpale

						Risk Evaluation and Response Definitions			
				H	High	Risk is unacceptable - move to reduce/eliminate exposure to hazard			
				M	Medium	Risk may be tolerable; more detailed review required; reduce risk to as low as reasonably practicle			
				L	Low	Risk is tolerable; continue to monitor and reduce risk to as low as reasonably practicle			
	Partial Risk (annual probability)								
Likelihood Descriptions	Indices		Probability Range						
Event typically occurs at least once per year	E	Almost certain	>0.9	M	M	H	H	H	H
Event typically occurs every few years	D	Very likely	0.1 to 0.9	M	M	H	H	H	H
Event expected to occur every 10 to 100 years	C	Likely	0.01 to 0.1	L	M	H	H	H	H
Event expected to occur every 100 to 1,000 years	B	Possible	0.001 to 0.01	L	L	M	H	H	H
Event expected to occur every 1,000 to 10,000 years	A	Unlikely	0.0001 to 0.001	L	L	L	M	H	H
	e (consequences)		^Indices	1	2	3	4	5	
			Low Hazard	Moderate Hazard	Significant Hazard	High Hazard	Catastrophic Hazard		
			Earthquake	Slight impact - Mecalli Intensity Scale (MIS): equal to or less than VII; recoverable in days to weeks; limited minor injuries	MIS VIII; Moderate injuries and/or personal hardship; recoverable withing weeks to months for communities with strong resilience capacity	MIS VIII; Serious injury and/or personal hardship; fatalities possible; recoverable over months to years for communities lacking resilience capacity	MIS IX-X; mass fatalities; serious personal hardship for survivors; recoverable over years for resilient communities; if settlement is in very close proximity to a significant fault settlement relocation to be considered	MIS XI-XII; Mass fatalities; most structures severely damaged or collapsed; recoverable over multiple years; if settlement is in very close proximity to a significant fault settlement relocation to be considered	
			Landslide	Small landslides that communities can provide appropriate mitigation for; injuries not likely; recoverable in days to weeks	Threat of medium magnitude losses to structures, infrastructure and/or agricultural land; mitigations viable & with local capacity; serious injuries and personal hardship possible if unaddressed; full recovery not likely	Threat of medium magnitude losses to structures, infrastructure and/or agricultural land; mitigations viable but may be beyond local capacity; serious injuries and fatatilies likely if unaddressed; full recovery not likely; at least partial relocation to be considered	Threat of high magnitude losses to structures, infrastructure and/or agricultural land; mitigation viability questionable; serious injuries and fatatilies highly likely if unaddressed, full recovery not likely; relocation of all or part of settlement to be considered	Threat of high magnitude losses to structures, infrastructure and agricultural land; mitigation not viable; serious injuries and fatatilies expected if unaddressed; not recoverable; relocation of settlement must be done	

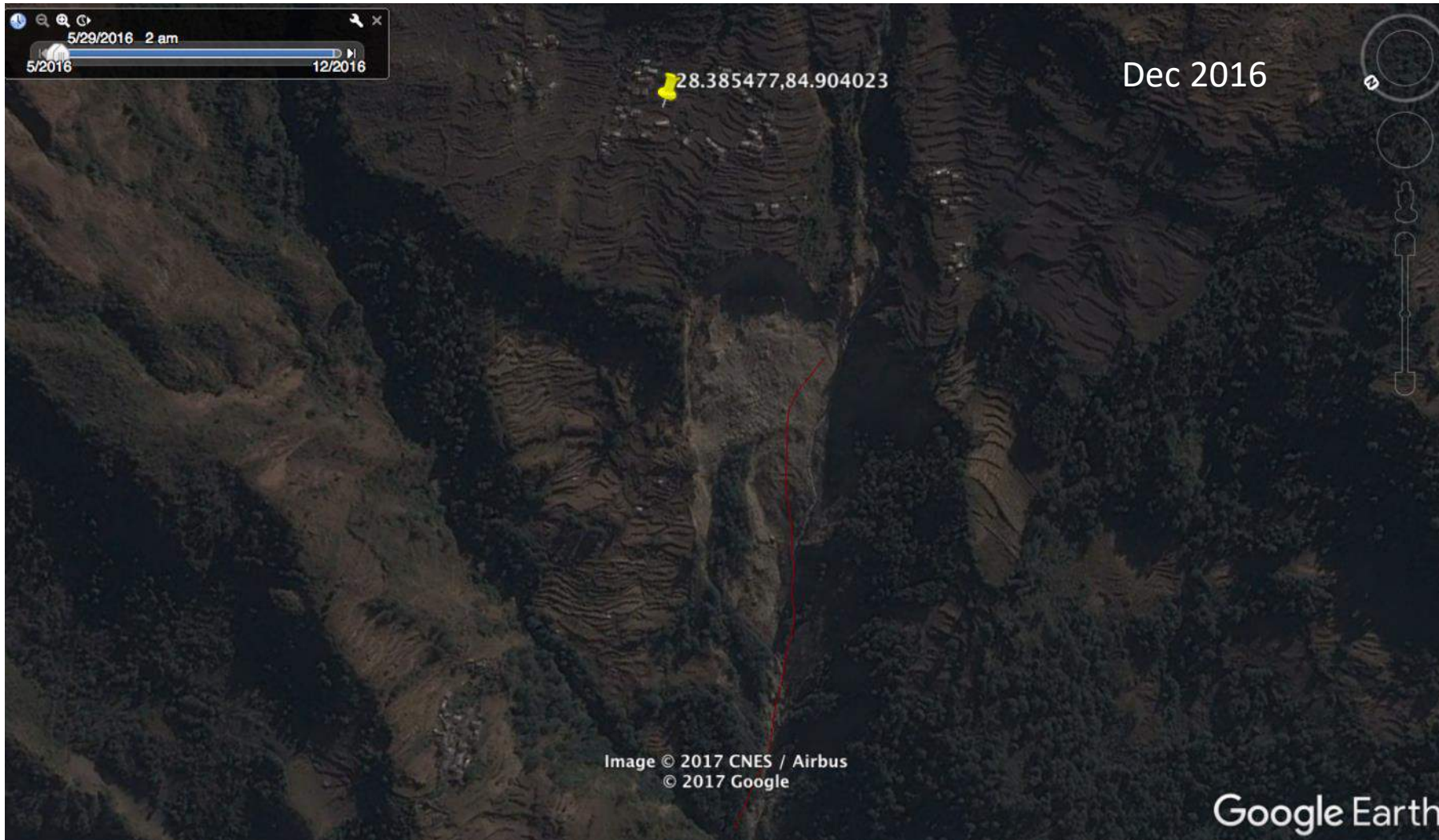
Gorkha: Sirdibas VDC – Dewal Gaun

NRA ASSESSMENT – Category 2: Drainage, bioengineering, slope trimming & retaining walls recommended



LANDSLIDE EXISTED BEFORE THE EARTHQUAKE – LOOKS TO HAVE DETERIORATED AS A RESULT & MAY BE GETTING WORSE

Gorkha: Sirdibas VDC – Dewal Gaun



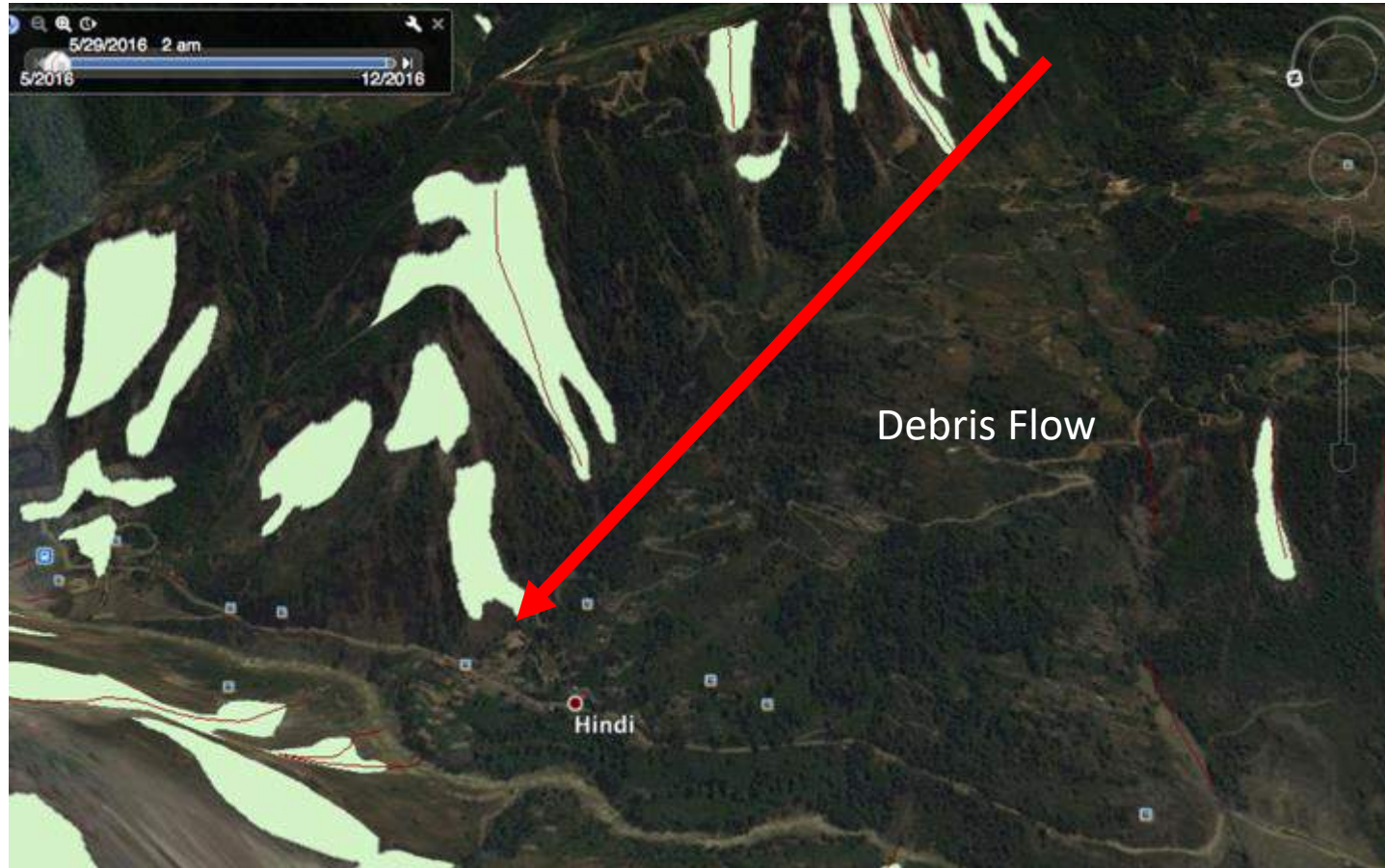
- 2 monsoons since earthquake.
- Crest of landslide appears to be migrating towards the village.
- Risk to community increasing with onset of 2017 monsoon?
- Major remediation works required – what is the cost, should this be Category 3?
- What is the GoN expectation on resilience – is that affordable?

Gorkha: Sirdibas VDC – Dewal Gaun

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Sindhupalchowk: Fulpingkatti VDC – Hindi Luksing (visited 5 March 2017)

NRA ASSESSMENT – Category 2 – Drainage, cascade construction, river training



Durham University data – mapped expansion of debris flow matched on ground



Sindhupalchowk: Fulpinkatti VDC – Hindi Luksing (visited 5 March 2017)

NRA ASSESSMENT – Category 2 – Drainage, cascade construction, river training



Debris flow behind school



Sindhupalchowk: Fulpinkatti VDC – Hindi Luksing (visited 5 March 2017)

NRA ASSESSMENT – Category 2 – Drainage, cascade construction, river training



- Is it feasible to utilize damaged school as debris trap – will require maintenance
- Should houses directly below debris flow be relocated reclassifying it as Category 3?

Sindhupalchowk: Fulpinkatti VDC – Hindi Luksing (visited 5 March 2017)

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4. NEXT STEPS

UNDERWAY

- Completion of geo-hazard survey
- Beneficiary survey (for Category 3 households that require relocation)

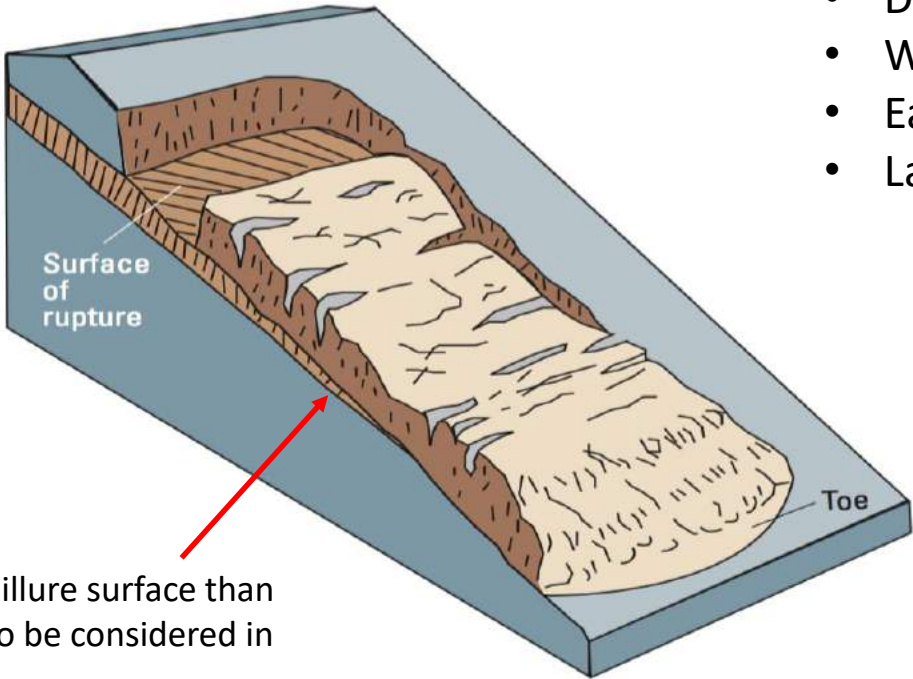
NEXT STEPS

1. **Category 3 Communities** - Engagement with affected communities regarding relocation options/process (Durable Solutions)
2. **Displaced Persons** - Further geo-hazard surveys of communities from which displaced persons have originated but were not included in the geo-hazard surveys undertaken by the DMG or NRA (NRA survey extension)
3. **Category 2 Communities** – Planning and costing for the undertaking of remedial works and implementation of risk management strategies. Establishing design criteria (design life, seismic and monsoon resilience), model remedial designs and standard details – incorporating work already done on landslide remediation in Nepal (NRA and others?)

4. TECHNICAL NOTE

REMEDIAL DESIGN CRITERIA – RESILIENCE & COST IMPACT

Translational failure in Rasuwa– soil and weathered rock (shallow circa 2 m)



Lower strength along failure surface than before failure – needs to be considered in the design

Level	Type	Cost	Resilience
1	Bioengineering	Low	No better than before failure (in this case)
2	Bioengineering & Drainage	Low - Med	Better– will prevent build up of water pressures in the soil mass. Limited resistance to peak monsoon conditions and earthquake loading.
3	Bioengineering, Drainage and Gabion Wall at Toe	Med	Better – structural support at toe to provide resistance to additional forces. Gabion wire may not last unless treated so could have limited design life – needs to be designed to take an agreed magnitude of earthquake loading. Wall foundations must be in stable ground.
4	Bioengineering, Drainage and Masonry or Concrete Wall at Toe	Med - High	Good – Masonry wall will have a longer life if designed and constructed properly – needs to be designed to take an agreed magnitude of earthquake loading. Wall drainage more on an issue

- Design life – 10, 20 or 50 Years?
- Water pressures?
- Earthquake loading?
- Lateral extent of remediation





QUESTIONS & DISCUSSION



Operational excellence for results that matter